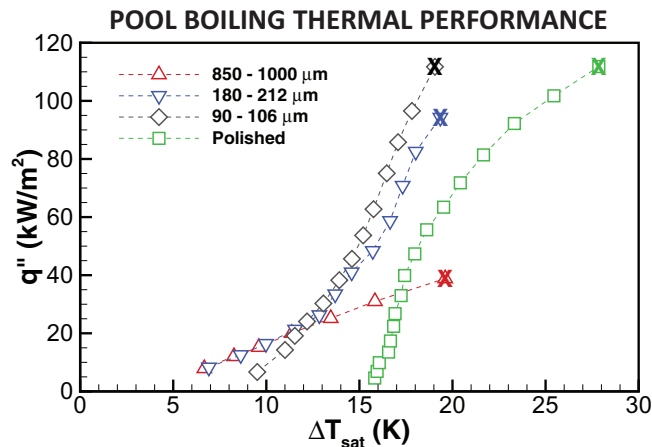
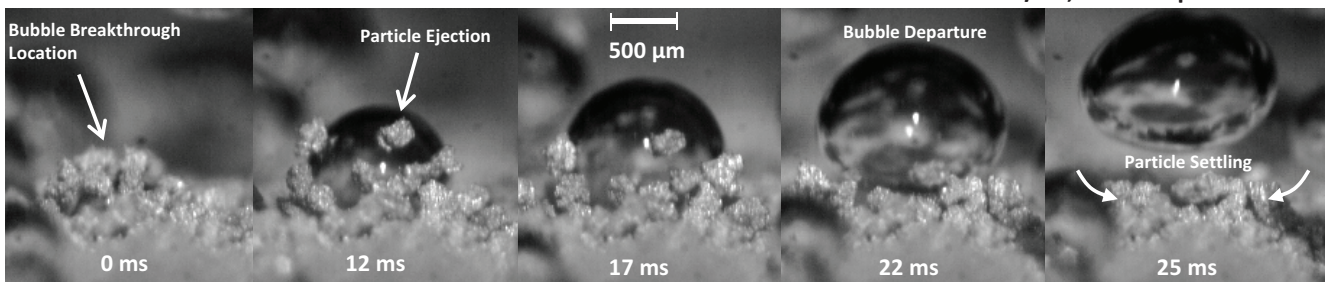




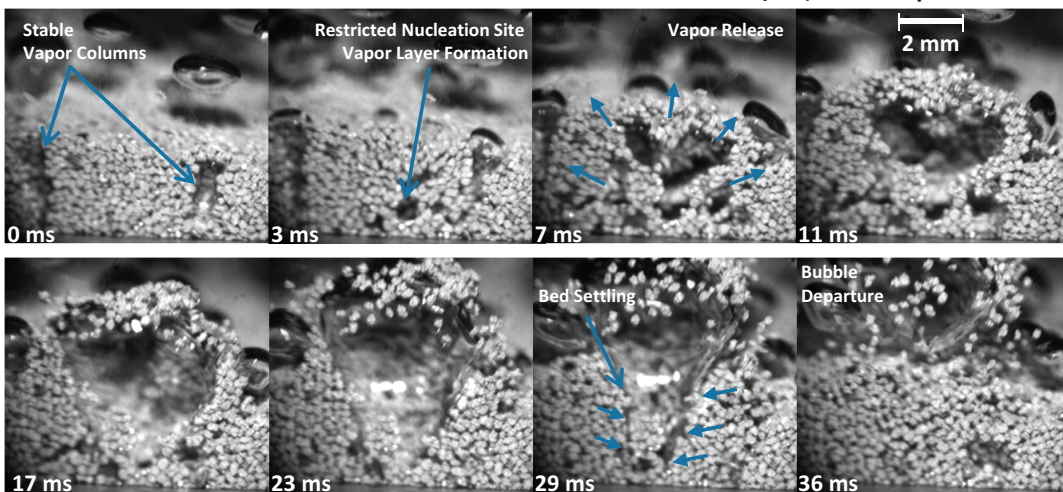
### VAPOR BUBBLE SURFACE DEPARTURE

4.0 kW/m<sup>2</sup>, 180-212  $\mu$ m Particles



### PASSIVE SURFACE DEFORMATION

5.9 kW/m<sup>2</sup>, 180-212  $\mu$ m Particles



## Pool Boiling Heat Transfer from Deformable Particulate Beds

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Pool boiling heat transfer is measured from a particulate bed of loose copper particles that are able to freely move on the polished copper surface beneath. The working fluid is FC-72. High-speed microscopic flow visualization at 3000 frames per second (Ultima APX, Photron) enables observation of vapor-particle interactions (VH-Z50L, Keyence). Free particle beds of 98, 196, and 925  $\mu$ m average particle diameter are directly compared to a baseline polished surface with average roughness  $< 0.1 \mu$ m (top). Layer thickness to particle diameter ratio is held constant. The largest particle size initially improves nucleate boiling performance, but bed deformation is not observed, and critical heat flux is reduced due to vapor trapping in the porous layer. In comparison, the smaller free particles retain nucleate boiling enhancement, but particles in the path of vapor departing from the surface are displaced and ejected into the liquid (middle). Visualizations confirm that passive deformation of the loose particulate bed eases vapor escape from the surface (bottom), enabling a 32% reduction in surface superheat up to the same critical heat flux as measured for the polished surface.